Pritika Kannapiran

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Education

University of Florida August 2025 – May 2027
Master of Science in Computer Science (MS), GPA: 3.88/4.0 *Gainesville, Florida*

Vellore Institute of Technology August 2021 – May 2025
Bachelor of Technology in Computer Science and Engineering (AI ML), GPA: 8.96/10 *Chennai, India*

Experience

Graduate Research Assistant Aug 2026 – Present
UF Health, Office of Research *Gainesville, Florida*

- Building an LLM-based auto-tagging pipeline with cron-scheduled inference, REST API and PostgreSQL

AI ML Intern May 2026 – August 2026
Honeywell Technologies *Atlanta, Georgia*

- Filed **3 invention disclosures**
- Deployed **Agentic SLM RAG** system with NLP to SQL OLAP querying on edge **NVIDIA Blackwell GPU**
- Built secrets management for etcd, MLflow, Milvus infra. Deployed **CI/CD** via **Kubernetes** on edge
- Boosted RAG retrieval accuracy **15% (65%→80%)** via RCA, retrieval redesign, and context engineering
- Architected a feedback-loop framework to improve model accuracy via real-time user feedback

Graduate Research Assistant September 2025 – May 2026
Intelligent Clinical Care Center - Grant funded by National Institutes of Health *Gainesville, Florida*

- **Automated 90%** of grant proposal pipeline for lab professionals using a LangGraph multi-agentic RAG system
- Researched **Model Context Protocol** (MCP) with **LLMs**, boosting the AI system performance by **70%**
- Involved in iterative testing lifecycle, achieving **85% user satisfaction** and identifying **30% key improvements**

R&D Intern - Agentic AI March 2025 – May 2025
AppViewX *Coimbatore, India*

- Contributed to the AI automation of 'Certificate Lifecycle Management', reducing manual processing time by **70%**
- Collaborated cross-functionally with front-end team to integrate 'Pages' framework, to render **14** UI components
- Developed an **Agentic AI** workflow that renders JSON into dynamic UI to improve operational processes by **80%**
- Experimented with **AWS Bedrock foundational models** and **Google Agent Development Kit**

Machine Learning Intern September 2023 – November 2023
AppViewX *Coimbatore, India*

- Developed and deployed the AppViewX Buddy, a real-time GenAI chat module for in-house queries
- Improved in-house communication efficiency by **75%**
- Optimized query processing and LLM prompt engineering, improving response accuracy by **33%**
- Integrated a CI/CD pipeline to monitor and validate vector store retrieval performance

Publications and Projects

VisionGraph-FPN | IEEE Xplore | *10.1109/ICICKE65317.2025.11136221* September 22, 2025

- Developed deep learning algorithm and a novel deep learning architecture using **Graph Neural Networks (GNNs)** to replace **FRCNN** backbone
- Built a PyTorch training pipeline for VisionGraph-FPN on medical data to improve feature extraction
- Accelerated the deep learning architecture on GPU using CUDA with an **IoU accuracy of 85.71%**

BERTaNER | IEEE Xplore | *10.1109/ICICKE65317.2025.11136417* September 22, 2025

- Developed an NLP model by fine-tuning a Tamil Dataset using **Python**, **PyTorch**, **TensorFlow** and **BERT**
- Leveraged **Adaptive Layerwise Learning Rate Decay** for **NER** in low-resource languages
- TamilBERT with A-LLRD achieves F1 scores of approximately **89.68%** on XTREME and **89.12%** on IndicGLUE

Technical Skills

Languages: Python, Java, JavaScript, C, C++
Tools & Tech: Git Version Control, Kubernetes, Docker, Jenkins, Flask, AWS, GCP, Azure, MongoDB, FAISS, Qdrant
Frameworks & Libraries: PyTorch, TensorFlow, LangChain, HuggingFace, FastAPI, Pandas, NLTK, NumPy